

computer_proj5

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1. Box-Muller Method

```
# Performing the Box-Muller method to generate normal distribution through sampling

start <- Sys.time()

# Set n = 5000 (for 5000 pairs and 10000 total)
n <- 5000

# Set up pairs of uniform RVs
U <- matrix(runif(n), ncol = 2)

# Calculate X1 and X2 through these equations:
#  $X_i = (-2\log(U_i))^{1/2} * \cos(2\pi U_{i+1})$ 
#  $X_{i+1} = (-2\log(U_i))^{1/2} * \sin(2\pi U_{i+1})$ 
X1 <- sqrt(-2*log(U[,1]))*cos(2*pi*U[,2])
X2 <- sqrt(-2*log(U[,1]))*sin(2*pi*U[,2])

# Combine pairs into X
X <- c(X1, X2)

# Complete Kolmogorov-Smirnov test on generated pairs
ks.test(X, "pnorm", mean = 0, sd = 1)
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: X
## D = 0.010173, p-value = 0.6788
## alternative hypothesis: two-sided
```

```
end <- Sys.time()
duration <- end - start
print(duration)
```

```
## Time difference of 0.0276649 secs
```

```
print(mean(X))
```

```
## [1] -0.009811239
```

```
print(sqrt(var(X)))
```

```
## [1] 0.988754
```

Remarks on Box-Muller Method

From the results of the Kolmogorov-Smirnov test, we fail to reject the null hypothesis with a very large p-value. We interpret this as $X_i \sim N(0, 1)$ as the null hypothesis in this test is that the data is distributed normally with $\mu = 0$ and $\sigma = 1$, and the large p-value does not provide enough evidence to reject the null hypothesis. The Box-Muller method successfully generates normally distributed RVs from originally uniformly distributed RVs

2. Marsaglia Polar Method

```
# Performing the Marsaglia Polar method to generate normal distribution through sampling
```

```
start <- Sys.time()
```

```
# Set n = 5000 (for 5000 pairs and 10000 total)
n <- 5000
```

```
# Establish empty X
X <- matrix(NA_real_, nrow = n, ncol = 2)
```

```
# Initialize counting index to 1
i <- 1
```

```
# Loop until have 5000 RV pairs
```

```
while (i<=n) {
  # Set up pairs of uniform RVs
  U <- runif(2, -1, 1)

  # Calculate R
  R <- U[1]^2 + U[2]^2

  # Check if R is in accepted region
  if (R>0 && R<=1) {
    # Calculate X_i and X_{i+1} by these equations
    # X_i = ((-2logR)/R)^(1/2) * U_i
    # and
    # x_{i+1} = ((-2logR)/R)^(1/2) * U_{i+1}
    X[i,1] <- sqrt(-2*log(R)/R)*U[1]
    X[i,2] <- sqrt(-2*log(R)/R)*U[2]
    i <- i+1
  }
}
```

```
# Complete Kolmogorov-Smirnov test on generated pairs
ks.test(X, "pnorm", mean = 0, sd = 1)
```

```
##  
## Asymptotic one-sample Kolmogorov-Smirnov test  
##  
## data: X  
## D = 0.0071386, p-value = 0.688  
## alternative hypothesis: two-sided
```

```
end <- Sys.time()  
duration <- end - start  
print(duration)
```

```
## Time difference of 0.1543891 secs
```

Remarks on Marsaglia Polar Method

Similarly to the previous method, the Marsaglia Polar Method seems to be doing a good job of generating the standard normal distribution and again is resulting in large p-values that suggest a failure to reject the null hypothesis, providing strong evidence of $X_i \sim N(0, 1)$. In terms of runtime efficiency, the Marsaglia Polar Method is more efficient due to it not using trigonometric functions, but for my solution this is not the case because I used a while loop and not a vectorized solution. In terms of generating uniform RVs though, the Box-Muller Method is more efficient as if you want to generate 5000 pairs, you only need to do just that, but with the Marsaglia Polar Method you may need to generate more pairs than that as a generated pair might not have an acceptable R value attached to them, and you have to regenerate.

3. Inverse CDF Method

```
# Using the Inverse CDF Method to generate normal distribution through sampling
```

```
start <- Sys.time()
```

```
# Set n = 10000
```

```
n <- 10000
```

```
# Generate uniform RVs
```

```
U <- runif(n)
```

```
# Obtain sign RVs from uniform RVs using indicator function
```

```
S <- U
```

```
S <- 2*as.numeric(S > (1/2)) - 1
```

```
# Generate another set of n uniform RVs
```

```
U_prime <- runif(n)
```

```
# Compute X via the formula:
```

```
#  $X_i = S_i * (\pi/8)^{1/2} * \log(1+U_i)/\log(1-U_i)$ 
```

```
X <- S*sqrt(pi/8)*log((1+U_prime)/(1-U_prime))
```

```
# Complete Kolmogorov-Smirnov test on generated pairs
```

```
ks.test(X, "pnorm", mean = 0, sd = 1)
```

```
##
```

```
## Asymptotic one-sample Kolmogorov-Smirnov test
```

```
##
```

```
## data: X
```

```
## D = 0.020599, p-value = 0.0004125
```

```
## alternative hypothesis: two-sided
```

```
end <- Sys.time()
```

```
duration <- end - start
```

```
print(duration)
```

```
## Time difference of 0.02443218 secs
```

```
print(mean(X))
```

```
## [1] 0.01572529
```

```
print(sqrt(var(X)))
```

```
## [1] 1.118586
```

Remarks on Inverse CDF Method

Unlike the other two methods, the small p-values generated from the Kolmogorov-Smirnov test suggest to reject the null hypothesis, meaning there is very strong evidence against $X_i \sim N(0, 1)$. The X RVs are not exactly Gaussian due to the fact that this is not an analytical solution to the inverse of the CDF of the Gaussin, it is numerical and an approximation by nature, as can be seen when I print the mean and variance at the end, they are close to 0 and 1 respectively but not quite close enough. If you wanted to have the X_i s to have an exact mean of -2, you can use the inverse CDF built into R and subtract 2, or if you'd like to approximate again, you can perform the same estimate as we did but subtract 2 from X.